

Projective Information Loss and Fibre Erasure in Admissible Non-Injective Projection

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Abstract

We establish a fibre-erasure theorem for the admissible coarse-graining hierarchy of the Cosmochrony spectral programme [3, 1]. The non-injective projection $\Pi : \Omega \rightarrow \mathcal{O}$ defines a residual fibre-information functional $I(c; \sigma(\ell))$, where c is a Weil-block fibre label and $\sigma_c(\ell)$ is the BFS capacity profile at coarse-graining depth ℓ . Conditional on the structural sufficiency hypothesis [H-suff] — that $\sigma(\ell)$ is a sufficient statistic for the hierarchy beyond ℓ — the data-processing inequality [9] implies that $I(c; \sigma(\ell))$ is non-increasing in ℓ : fibre information is erased, not created, under admissible coarse-graining. This result is *not* a c -theorem; it does not count effective degrees of freedom. It is a quantitative realisation of the ENI no-go theorem [3]: non-injectivity of Π forces projective information loss, and the data-processing inequality makes this loss monotone and measurable. The inter-sector variance $\text{Var}_c(\sigma_c(\ell))$ serves as a computable proxy; its restriction to $\text{SU}(3)$ colour-triplets connects directly to hypothesis [H-color] [8, 7].

Keywords. non-injective projection; fibre erasure; admissible coarse-graining; data-processing inequality; information monotone; BFS capacity; spectral admissibility; projective information loss; Weil representation; Heisenberg group; [H-color]

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1 Introduction

1.1 Contributions and non-claims

This paper establishes, conditional on hypotheses [H-suff] and [H-sg] (Section 2.4):

- (i) a monotone information functional $I(c; \sigma(\ell))$ for the BFS coarse-graining hierarchy (Theorem 4.3);
- (ii) its non-increase under depth-increment, via the data-processing inequality;
- (iii) the identification of $I(c; \sigma(\ell)) = 0$ as the fibre-erasure fixed-point condition (Corollary 4.4).

This paper does *not* establish:

- (i) a c -theorem or analogue (no counting of effective degrees of freedom; see open problem [O-2]);
- (ii) hypothesis [H-suff] itself (stated and motivated; see open problem [O-1]);
- (iii) the semigroup hypothesis [H-sg] (stated and motivated; see open problem [O-1]).

Strict disclaimer. This result should not be called a c -theorem. It is a *fibre-erasure theorem*: the monotone measures the loss of residual fibre-label information under admissible coarse-graining. A c -theorem-type statement would require an additional counting principle for effective degrees of freedom, which is not established here and is listed as open problem [O-2] in Section 6.

ENI versus monotonicity. ENI [3] establishes that information is structurally lost under non-injective projection, but does not by itself imply any monotone hierarchy over the coarse-graining scale ℓ . The monotonicity of Theorem 4.3 requires the additional hierarchical structure encoded in [H-suff] and [H-sg]: these are the informational content this paper contributes beyond ENI.

1.2 Programme placement

2 Framework

2.1 Non-injective projection and fibre structure

We recall the setting of [3, 1].

Definition 2.1 (Non-injective projection and fibre). A *projective description* is a surjective map $\Pi : \Omega \rightarrow \mathcal{O}$ such that $\mathcal{O}$ carries no microstate-resolving structure independent of Π . The *fibre* over $o \in \mathcal{O}$ is $\Pi^{-1}(o) = \{\omega \in \Omega : \Pi(\omega) = o\}$. The *projection entropy* is $S_\Pi = \mathbb{E}_o[\log |\Pi^{-1}(o)|] \geq 0$, with $S_\Pi > 0$ whenever Π is non-injective (ENI Theorem 2.1, [3]).

In the spectral realisation [1], the fibre structure is implemented via the Weil representation V_ρ of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ for a prime q . The minimal non-injectivity is the parity involution $c \leftrightarrow q - c$: Weil blocks ρ_c and ρ_{q-c} map to the same effective observable class. The *fibre label* $c \in \{1, \dots, (q-1)/2\}$ indexes blocks within an equivalence class under Π .

2.2 The BFS coarse-graining hierarchy

Definition 2.2 (BFS capacity profile). For a prime q , a fibre label c , and BFS depth $\ell \in \mathbb{N}$, the *BFS capacity profile* $\sigma_c(\ell) \in [0, 1]$ is the normalised residual projective capacity of block c at depth ℓ along the Heisenberg Cayley graph BFS cascade [5, 6]. The *pair observable* is $\sigma_{\text{pair}}(\ell) = \sigma_c(\ell) \cdot \sigma_{q-c}(\ell)$ on conjugate pair $(c, q - c)$.

2.3 Three-axis separation

Axis	Parameter	Role
Coarse-graining / resolution	ℓ (BFS depth/window)	coarse-graining hierarchy; carries $I(c; \sigma(\ell))$; this paper
Thermodynamic / continuum	$q \rightarrow \infty$	Continuum limit; Conjecture 4.7 of [2]
Temporal / cascade	n (cascade step)	Projective time; carries $I(n)$ of [4]

Table 1: The three distinct axes of the observable hierarchy. The coarse-graining axis ℓ is the support of this paper. The temporal axis n is the support of PTO [4]. The continuum axis $q \rightarrow \infty$ is a separate thermodynamic limit.

Remark 2.3 (Cascade and coarse-graining are orthogonal). The admissible cascade $U_0 \rightarrow U_1 \rightarrow \dots$ defines the temporal axis n and is *not* an RG flow. RG-like behaviour is carried by Π and the semigroup $\{G_{\ell' \leftarrow \ell}\}$ on the ℓ -axis. The cascade *enlarges* the admissible space as temporal ordering advances; an RG flow toward the IR *reduces* effective degrees of freedom. These are distinct phenomena on orthogonal axes.

2.4 The coarse-graining semigroup

Definition 2.4 (Coarse-graining channel). For $\ell' > \ell$, the *coarse-graining channel* $G_{\ell' \leftarrow \ell} : \sigma(\ell) \mapsto \sigma(\ell')$ is the map sending the BFS capacity profile at depth ℓ to the profile at depth ℓ' .

Hypothesis 2.5 ([H-sg]: semigroup property). The coarse-graining channels satisfy

$$G_{\ell_3 \leftarrow \ell_2} \circ G_{\ell_2 \leftarrow \ell_1} = G_{\ell_3 \leftarrow \ell_1} \quad \text{for all } \ell_1 < \ell_2 < \ell_3. \quad (1)$$

2.5 Probability measure on fibre labels

Definition 2.6 (Fibre-label measure). Let μ be a probability measure on $\{c : 1 \leq c \leq (q-1)/2\}$. Two canonical choices are used:

- (i) *Uniform*: $\mu(c) = 2/(q-1)$ for all c .
- (ii) *Triplet-uniform* (for $q \equiv 1 \pmod{3}$): μ uniform on $\{c, \omega c, \omega^2 c\}$, where ω is a primitive cube root of unity modulo q ; used in the [H-color] context [8, 7].

The choice of μ does not affect the monotonicity of $I(c; \sigma(\ell))$ (which holds for any fixed measure by DPI), but affects its value and the interpretation of the fixed point.

3 Structural Sufficiency and the Markov Property

Hypothesis 3.1 ([H-suff]: structural sufficiency). For all $\ell' > \ell$, the coarse-graining channel $G_{\ell' \leftarrow \ell}$ is conditionally independent of the fibre label c given $\sigma(\ell)$:

$$c \longrightarrow \sigma(\ell) \longrightarrow \sigma(\ell'). \quad (2)$$

That is, $\sigma(\ell)$ is a *sufficient statistic* for $\sigma(\ell')$ with respect to the fibre label c . The term “sufficient” is used in the information-theoretic sense: all information in $\sigma(\ell)$ that is predictive of $\sigma(\ell')$ is already fully encoded; no additional knowledge of c reduces the uncertainty in $\sigma(\ell')$ beyond what $\sigma(\ell)$ already captures.

Remark 3.2 (Interpretive status of [H-suff]). Hypothesis [H-suff] is not introduced solely to enable the DPI. It is the informational reformulation of an effective closure property already implicit throughout the spectral admissibility programme: if [H-suff] failed — if BFS dynamics at depth ℓ' re-accessed label information not encoded in $\sigma(\ell)$ — then $\sigma_c(\ell)$ would not constitute stable effective observables under Π , and the convergence $\sigma_c \rightarrow \sigma^*$, the plateau structure across $q \in \{61, 151, 211\}$, and the [H-color] campaign [8, 7] would be structurally very difficult to account for. More precisely, [H-suff] is the condition that the BFS observable $\sigma(\ell)$ constitutes a *complete summary* of the fibre structure at resolution ℓ : the projection Π has already collapsed the relevant fibre information at each level of the hierarchy, and no deeper BFS step retrieves it. This is exactly what the non-injectivity of Π (ENI [3]) demands at the effective level: fibre information is not gradually re-injected as coarse-graining proceeds, but is collapsed once and for all by Π . In this sense, [H-suff] makes explicit, in information-theoretic language, a structural commitment already implicit in the programme's use of $\sigma_c(\ell)$ as an effective observable.

Remark 3.3 (Status of [H-suff]). Hypothesis [H-suff] is strongly motivated by Remark 3.2 and is compatible with all known spectral results. It is not yet proved; its proof is listed as open problem [O-1].

Remark 3.4 (Why [H-suff] is load-bearing). If hypothesis [H-suff] fails for some pair (ℓ, ℓ') — that is, if $\sigma(\ell')$ depends on c beyond its dependence on $\sigma(\ell)$ — then no structural information monotone of the form $I(c; \sigma(\ell')) \leq I(c; \sigma(\ell))$ can be guaranteed. If $p(\sigma(\ell') \mid \sigma(\ell), c) \neq p(\sigma(\ell') \mid \sigma(\ell))$ for some c , then the chain $c \rightarrow \sigma(\ell) \rightarrow \sigma(\ell')$ is not Markov, and the data-processing inequality does not apply. This is not a limitation of the proof technique; it reflects the genuine logical dependence of the monotonicity on the Markov structure.

4 The Fibre-Erasure Theorem

4.1 The information functional

Definition 4.1 (Residual fibre information). The *residual fibre information* at depth ℓ is

$$I(c; \sigma(\ell)) = H(c) - H(c \mid \sigma(\ell)) = H(\sigma(\ell)) - H(\sigma(\ell) \mid c), \quad (3)$$

where $H(\cdot)$ denotes Shannon entropy with respect to measure μ .

Remark 4.2 (Physical interpretation). $I(c; \sigma(\ell)) = 0$ if and only if $\sigma(\ell) \perp c$: the BFS profile carries no information about the microscopic fibre representative. This is the operational statement of fibre erasure under Π .

4.2 Main theorem

Theorem 4.3 (Fibre-Erasure Theorem). *Assume hypotheses [H-suff] (Hypothesis 3.1) and [H-sg] (Hypothesis 2.5). Then for all $\ell' > \ell \geq 0$:*

$$I(c; \sigma(\ell')) \leq I(c; \sigma(\ell)). \quad (4)$$

In particular, $I(c; \sigma(\ell))$ is non-increasing in ℓ and converges to a limit $I_\infty \geq 0$.

Proof sketch. By [H-suff], the triple $(c, \sigma(\ell), \sigma(\ell'))$ forms a Markov chain (2). The data-processing inequality [9] applied to this chain gives (4) directly. Composability [H-sg] extends the inequality from $\ell' = \ell + 1$ to all $\ell' > \ell$. $\square$

Corollary 4.4 (Fibre-erasure fixed point). *Under the hypotheses of Theorem 4.3, $I(c; \sigma(\ell)) = 0$ if and only if $\sigma(\ell) \perp c$. This is the fibre-erasure fixed point: the coarse-graining has fully collapsed the fibre-label information.*

Explicit disclaimer. Theorem 4.3 is *not* a c -theorem. At the fibre-erasure fixed point, $I_\infty = 0$ does not count any effective degrees of freedom. A c -theorem analog requires a separate counting functional; see open problem [O-2].

4.3 Information decomposition and programme connections

By the chain rule for entropy:

$$H(\sigma(\ell)) = H(\sigma(\ell) \mid c) + I(c; \sigma(\ell)). \quad (5)$$

Here $I(c; \sigma(\ell))$ is the fibre-erasure functional (Theorem 4.3), monotone non-increasing in ℓ ; $H(\sigma(\ell) \mid c)$ is the intrinsic entropy of the observable conditioned on the fibre label, and is the natural candidate for a counting monotone [O-2].

Remark 4.5 (Connection to PTO and axis orthogonality). The temporal monotone $I(n) = \sigma_{\text{pair}}(0) - \sigma_{\text{pair}}(n)$ of [4] lives on the temporal axis n (Table 1). The present $I(c; \sigma(\ell))$ lives on the coarse-graining axis ℓ . They measure distinct phenomena on orthogonal axes and should not be conflated.

5 Numerical Proxies and Spectral Programme Connections

5.1 Inter-sector variance as proxy

Definition 5.1 (Inter-sector variance). The *inter-sector variance* at depth ℓ is

$$\mathcal{V}(\ell) = \text{Var}_c(\sigma_c(\ell)) = \mathbb{E}_c[\sigma_c(\ell)^2] - (\mathbb{E}_c[\sigma_c(\ell)])^2. \quad (6)$$

$\mathcal{V}(\ell) = 0$ is a necessary condition for $I(c; \sigma(\ell)) = 0$, but $\mathcal{V}$ is *not* guaranteed to be monotone in ℓ . Non-monotone behaviour of $\mathcal{V}$ would not contradict Theorem 4.3; it would indicate that variance is an imperfect surrogate for $I(c; \sigma(\ell))$.

5.2 Triplet variance and [H-color]

For $q \equiv 1 \pmod{3}$, the *triplet-restricted variance* is

$$\mathcal{V}_{\text{triplet}}(\ell) = \text{Var}(\sigma_c(\ell), \sigma_{\omega c}(\ell), \sigma_{\omega^2 c}(\ell)), \quad (7)$$

averaged over colour triplets.

Remark 5.2 ([H-color] as a fibre-erasure statement). Hypothesis [H-color] of [8, 7] asserts $\sigma_c(\ell) = \sigma_{\omega c}(\ell) = \sigma_{\omega^2 c}(\ell)$ for $q \equiv 1 \pmod{3}$, which is precisely $\mathcal{V}_{\text{triplet}}(\ell) = 0$ and, under [H-suff], $I_{\text{triplet}}(c; \sigma(\ell)) = 0$. The fibre-erasure theorem provides the information-theoretic framework within which [H-color] is a special case: the colour label becomes unobservable at the effective level. The triplet-restricted variance is already computable from the $q \in \{61, 151, 211\}$ runs of O32 [7].

5.3 Numerical programme

6 Open Problems

[O-1] Proof of [H-suff] and [H-sg].

Show that the BFS depth-increment map $G_{\ell+1 \leftarrow \ell}$ is a Markov kernel on σ : the transition $p(\sigma(\ell+1) \mid \sigma(\ell), c)$ is independent of c . This is the load-bearing prerequisite for Theorem 4.3. A first numerical test: does conditioning on c in addition to $\sigma(\ell)$ reduce prediction error for $\sigma(\ell+1)$?

[O-2] c -theorem analog via $H(\sigma(\ell))$.

The decomposition (5) identifies $H(\sigma(\ell) \mid c) = H(\sigma(\ell)) - I(c; \sigma(\ell))$ as a candidate counting monotone. Establish its monotonicity in ℓ and verify that its fixed-point value counts effective degrees of freedom. This would upgrade the fibre-erasure theorem to a genuine c -theorem analog.

[O-3] Temporal causal structure and emergent light cone.

Define the temporal mutual information $I_{\text{temp}}(n, k) = I(U_n; U_{n+k})$ and the transfer entropy $T(U_n \rightarrow U_{n+k}) = I(U_{n+k}; U_n \mid U_{n-1}, \dots, U_0)$. Their decay in k would yield an operational definition of the emergent forward light cone on the temporal axis n , orthogonal to the present paper.

[O-4] Axis disambiguation: ℓ -limit versus $q \rightarrow \infty$.

Determine whether the fibre-erasure fixed point σ^* is approached as a limit in ℓ at fixed q (a Wilsonian fixed point of the semigroup) or only in the limit $q \rightarrow \infty$ (a thermodynamic/continuum limit; see Conjecture 4.7 of [2]).

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